

TIN DANG HUYNH TRUNG

Senior Software Engineer – Backend, AI Platforms & Distributed Systems

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Senior software engineer with eight years building and operating production backend systems in Python, Rust and TypeScript across fintech, conversational AI, computer vision and data platforms. Sole architect on several greenfield platforms and technical lead on teams of 5–14. I prefer correctness enforced by structure (ledger invariants, fail-closed multi-tenancy, idempotent money movement), performance that is measured with the methodology published, and shipping: packages on PyPI, crates.io and npm with 100+ tagged releases between them.

PROFESSIONAL EXPERIENCE

Senior Software Engineer / Technical Lead

Trustify Technology

Jan 2024 – Present

Ho Chi Minh City, Vietnam

Consultancy delivering fintech, real-estate and intelligent-transport platforms in Vietnam and Myanmar (Yoma Group, Yoma Bank, Wave Money, Trustonic). Architect and lead engineer on client rebuilds; author of the company engineering career ladder, its spec-driven development method and the Azure Databricks lakehouse proposal for VinaCapital.

- **Ledger-first estate-finance platform for Yoma Group (sole engineer, 1,500+ commits).** Rebuilt a live property-management and treasury system from PHP/Laravel onto FastAPI, Python 3.12 and async SQLAlchemy 2 with schema-per-tenant PostgreSQL: 32 bounded modules including customer cutover, transactional outbox, dead-letter queue and reconciliation; 980+ test files behind an 80% coverage floor on the money core. Graded 235 legacy business rules across 39 modules, cited to file and line on both sides, before any customer moved.
- **Payments and PCI boundary.** Isolated cardholder data into its own billing service, database and network, with the boundary enforced by a static import ban guarded in CI; integrated 2C2P card payments with idempotent callbacks and three-way reconciliation, and built a Wave Money wallet prototype on the same gateway. Ledger balance and row immutability are enforced by database triggers; a single fixed-point money type fails closed on floats.
- **Urban intelligent-traffic platform for Ho Chi Minh City (architect and sole engineer, 840 commits).** Three-tier design for 400+ cameras: Jetson edge nodes on K3s with ONVIF/RTSP ingest and H.265 filtering (75% less bandwidth), YOLO inference on TensorRT/Triton (P95 85 ms), Kafka with mTLS and per-tenant ACLs, PyFlink exactly-once ingestion measured at 71K events/s, GDPR-pseudonymised plate data. 12,700+ tests, ArgoCD edge fleet, 69-diagram C4 architecture set.
- **Yoma Bank and Star City.** Built Yoma Bank's microfinance lending MVP in Java 21 / Spring Boot 3.3 (16 Gradle modules, hexagonal, jOOQ, Kafka, Flyway) and led its bilingual culture-survey build; owned the money, auth and payment seams on the Star City Living resident platform with a nine-person team, closing auth-enumeration, TOCTOU and privilege-escalation findings.
- **Trustonic device-management migration.** Tuned the PySpark / AWS Glue / Airflow jobs moving customers between platform generations (vectorised batch validation, broadcast joins, keyed DynamoDB reads, measured Redis partitioning) and owned the per-customer rollback job across PostgreSQL and DocumentDB.

Founder & Principal Engineer

Pilot Space (open-source AI infrastructure)

Jan 2026 – Present

Remote

- **Moon.** Redis-compatible in-memory store written from scratch in Rust: RESP2/RESP3, 250+ commands, replication, io_uring runtime, vector, BM25 and graph search, MVCC. 30 releases to v0.8.9 on crates.io and PyPI; measured 1.62× GET and 1.32× SET against Redis 7.0 with 8–22% less memory per key, losing rows published. Fuzzing on every PR, loom and crash-injection recovery tests, CI-gated unsafe policy.
- **Lunaris and Helios.** Agent-memory engine on Moon with p50 recall of 19–22 ms at 100K documents per scope and a CI perf gate that fails on 5% regression (crates.io, npm). Helios is a 48-crate Rust agent harness: 12 provider backends, native MCP client and server, WASM-sandboxed extensions, explicit timeout, retry and circuit-breaker policy on every outbound call.
- **Hydroa and Pilot Space.** Multi-tenant LLM gateway (Envoy ext_authz in front of FastAPI, 6 providers, per-tenant and per-provider circuit breakers, BYOK keys encrypted at rest, cost ledger reconciled against provider billing, OpenTelemetry) with 4,600 tests and CI sharded from 80 minutes to 15; an AI-augmented SDLC platform on the same stack with PostgreSQL row-level security on every table and 5,300 tests.
- **ADD.** Published spec-first, red/green AI-driven development method (@pilotspace/add on npm, pilotspace-add on PyPI; v3.6, 36 releases, about 1.2K downloads a month); the working method on the client platforms above.

Technical Lead

ClickAI Viet Nam

📅 Jun 2024 – Oct 2025 (concurrent)

📍 Ho Chi Minh City, Vietnam

Conversational AI platform (clickai.vn) with multi-modal chat, image and audio generation and community features, serving 40K+ users across Southeast Asia. Top committer (2,300+ commits across 14 repositories) on a 12–14 person product team.

- **Billing engine.** Sole author of wallets, plans, promotions, referral commissions and a month-partitioned PostgreSQL transaction ledger metering every LLM call. A reconciliation check I wrote found provider billing had drifted 30%+ from metered cost; fixed the ledger model and indexes, added source-event idempotency and promoted the check to a standing job.
- **Model platform.** Maintained a customised Dify fork for 17 months (640+ commits, 11.7K lines of new API surface, 20 model providers plus a custom one, vector-memory layer, reasoning-model support) and a 42K-line generation service fanning out to 7 backends (ComfyUI, Replicate, fal.ai, ElevenLabs, OpenAI) behind circuit breakers, retries and webhook reconciliation. Policy-based model routing cut LLM API spend by 20%.
- **Production ownership.** Authored the 13-service Helm umbrella on Kubernetes (Azure) with cert-manager, probes and autoscaling, Weaviate and Qdrant vector stores; moved PostgreSQL to a 3-node Patroni HA cluster after a saturation incident; sub-2s responses at 99.9% uptime over 1K+ concurrent sessions.
- **Identity and web.** OAuth (Google, Microsoft), TOTP, refresh-token rotation and a sync-to-async SQLAlchemy migration of the identity service; 400 commits on the 145K-line Next.js 14 client (SSE streaming, organisation workspaces, RBAC, i18n).

Earlier Experience (2018 – 2024)

- **Senior Data Engineer**, Asia Commercial Bank (ACB) • Aug 2023 – Jun 2024 — Enterprise data lake on Cloudera at 5M+ daily transactions; a Dagster and PySpark data-quality framework across six quality dimensions cut manual verification by 80%; FastAPI services consumed by 100+ internal applications.
- **Software Data Engineer**, Vietmoney Corp • Mar 2021 – Aug 2023 — Fintech appraisal and lending platform at 10M+ daily transactions on GCP, 99.9% uptime. Price forecasting at 96% accuracy cut appraisal from 15 minutes to 5 seconds; Vietnamese NER at 96% F1; a distributed Scrappy, Pub/Sub and Kubernetes crawler doubled throughput across 50+ sources; BigQuery Kimball warehouse over 1TB+.
- **Software Developer**, Dai Phat Solutions • Aug 2019 – Mar 2021 — CCTV analytics at 15 FPS for 20+ clients cut reported incidents by 40%; face recognition at 92% accuracy over 1,000+ employees; published `dpsutil` to PyPI (44 releases: mmap vector pools, Blosc compression, Kafka, TurboJPEG and FFmpeg media I/O).
- **Full-stack Developer**, LogiGear Corporation • Jun 2018 – Mar 2019 — Web3 wallet, smart-contract and liquidity-pool management on Ethereum with MetaMask; Node.js and React.

OPEN SOURCE

- **Moon**, **Lunaris** and **ADD** at github.com/pilotspace (see above). **yowo** on PyPI: multi-backend YOLO inference and export (TensorRT, ONNX Runtime, CoreML, OpenVINO) with ByteTrack and CLIP-ReID tracking, 2,100+ tests. **fastapi-multi-tenant** (429 tests), **rjson** (Rust-backed JSON for Python), **dpsutil**, **face-anti-spoofing** (founded at Vietmoney).

TECHNICAL SKILLS

Languages: Python (typed, async), Rust (Tokio, Axum, SQLx), TypeScript / Node.js, SQL / PL/pgSQL, Java (Spring Boot), Bash

Backend & architecture: FastAPI, SQLAlchemy 2, Pydantic, Next.js / React, clean and hexagonal architecture,DDD, CQRS, event sourcing, multi-tenancy, outbox, idempotency, PCI boundaries, OAuth / OIDC

AI / ML: PyTorch, YOLO, TensorRT, ONNX Runtime, Triton, OpenCV, LLM routing and gateways, RAG, agent memory, MCP, evals, guardrails

Data & streaming: Kafka, PyFlink, PySpark, AWS Glue, Airflow, Dagster, BigQuery, Databricks / Delta Lake, DuckDB

Datastores: PostgreSQL (Patroni HA, RLS, partitioning), Redis, MongoDB / DocumentDB, DynamoDB, Weaviate, Qdrant

Cloud & DevOps: Docker, Kubernetes, K3s, Helm, ArgoCD, GitHub Actions, Azure, AWS (ECS, Glue, S3, RDS), GCP, Prometheus, OpenTelemetry

EDUCATION, TEACHING & CERTIFICATIONS

- **BSc Computer Science (High-Quality Program)**, Ton Duc Thang University, 2015 – 2019, GPA 3.2. Research Intern, NewAI Lab TDTU (2018).
- **Teaching:** Instructor, Design Patterns course, Ton Duc Thang University (2022); Data Analysis Mentor, MindX Technology School (2023); Speaker, AI Workshop, FPT University (2025).
- **Certifications:** Google Cloud Technical Series – AI Agents (2025); AWS Machine Learning Foundations, Udacity (2022); Agile Software Development with Scrum, Axon Active (2017). **Languages:** English, Vietnamese.